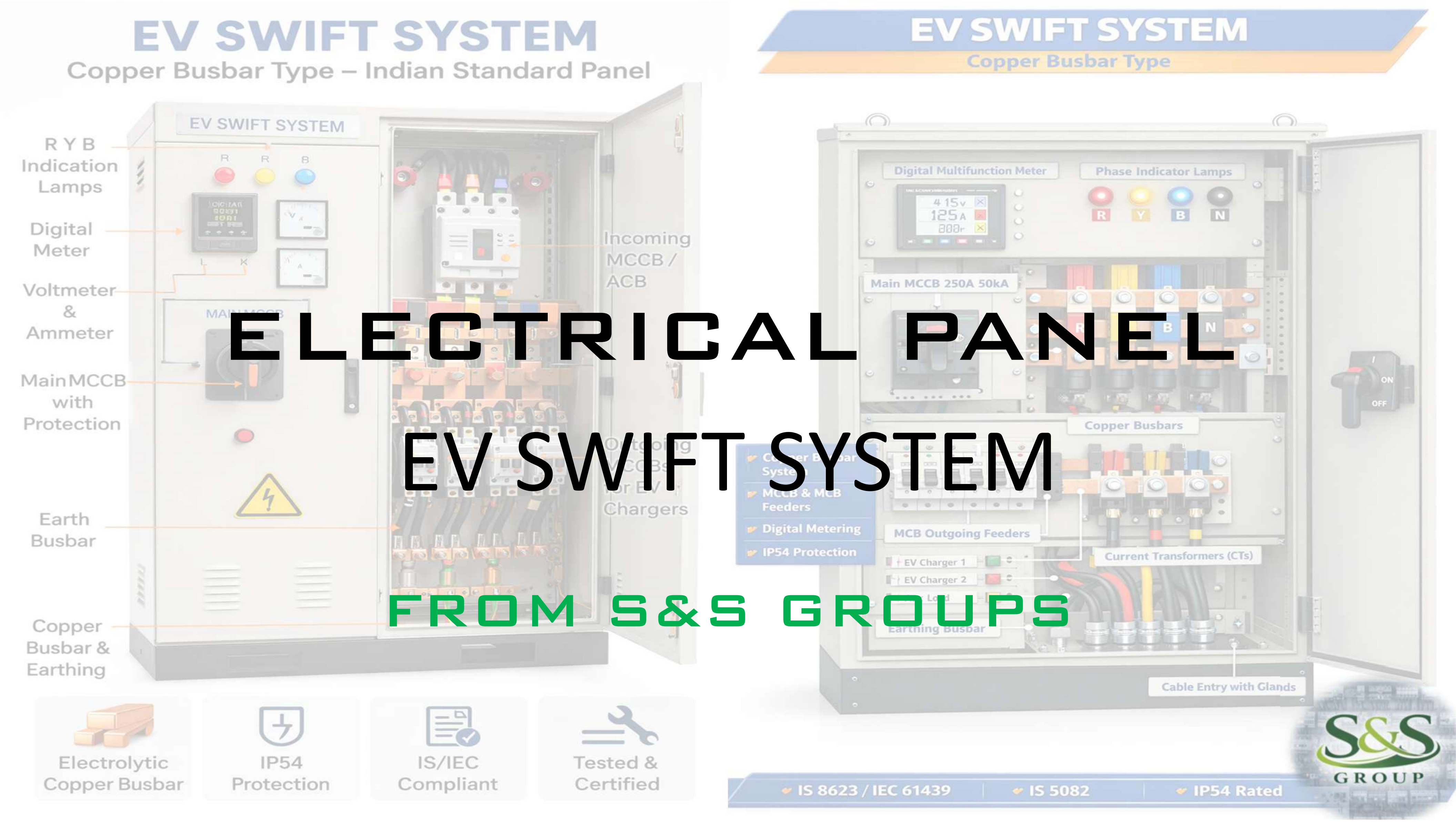
ELECTRICAL PANEL

EV SWIFT SYSTEM

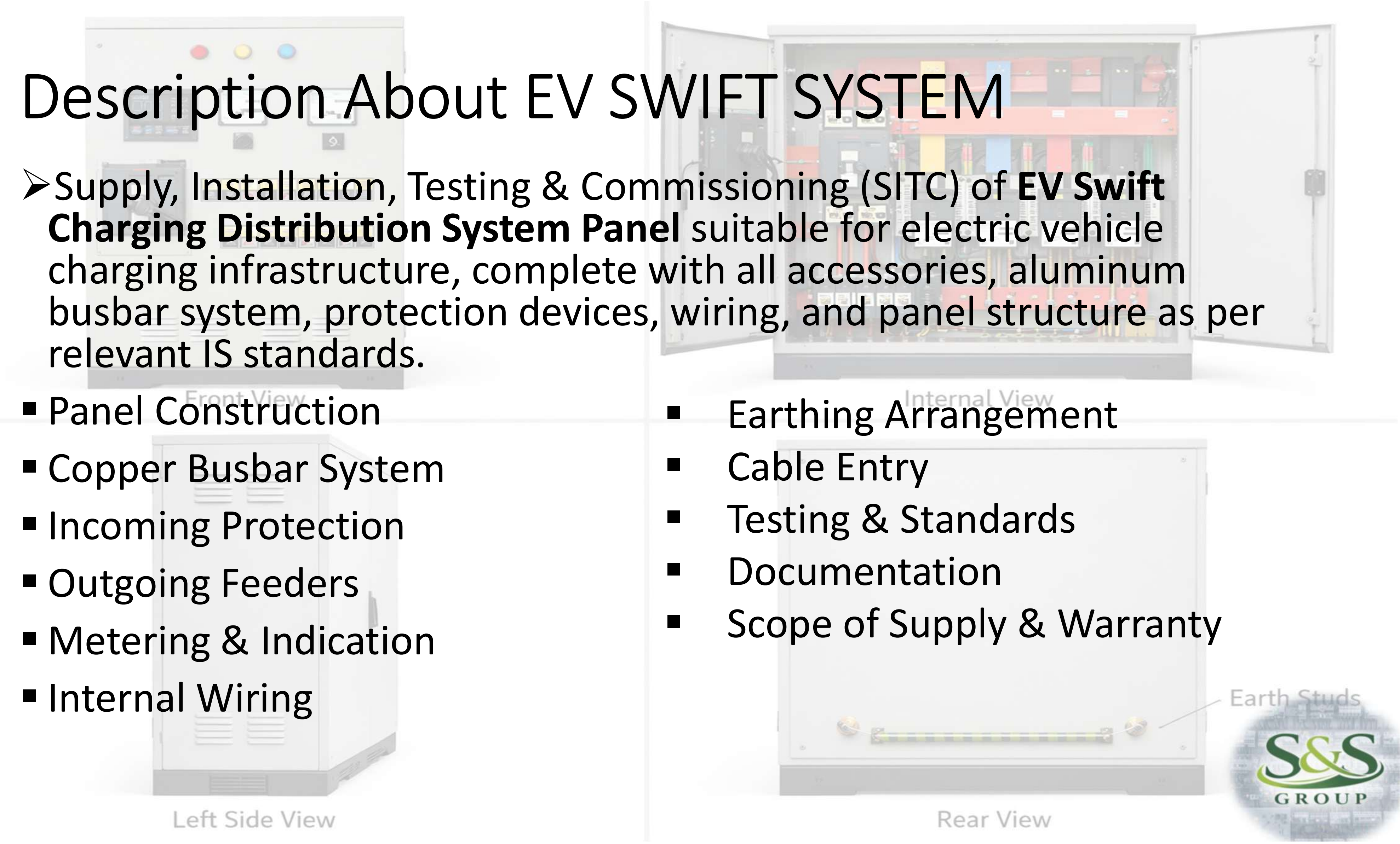


FROM S&S GROUPS

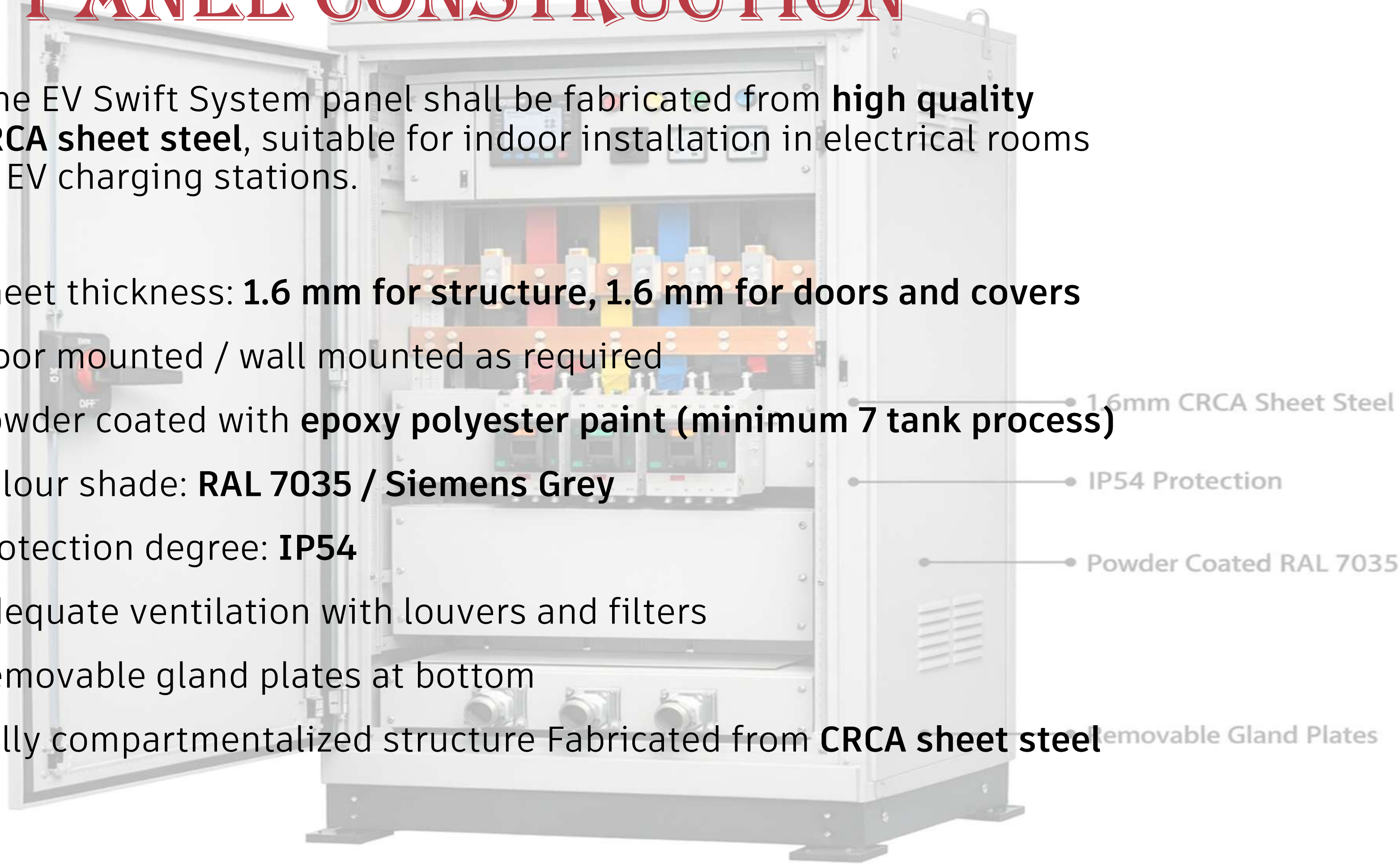
Description About EV SWIFT SYSTEM

➢ Supply, Installation, Testing & Commissioning (SITC) of **EV Swift Charging Distribution System Panel** suitable for electric vehicle charging infrastructure, complete with all accessories, aluminum busbar system, protection devices, wiring, and panel structure as per relevant IS standards.

- Panel Construction
- Copper Busbar System
- Incoming Protection
- Outgoing Feeders
- Metering & Indication
- Internal Wiring
- Earthing Arrangement
- Cable Entry
- Testing & Standards
- Documentation
- Scope of Supply & Warranty



1. PANEL CONSTRUCTION



❑ The EV Swift System panel shall be fabricated from **high quality CRCA sheet steel**, suitable for indoor installation in electrical rooms or EV charging stations.

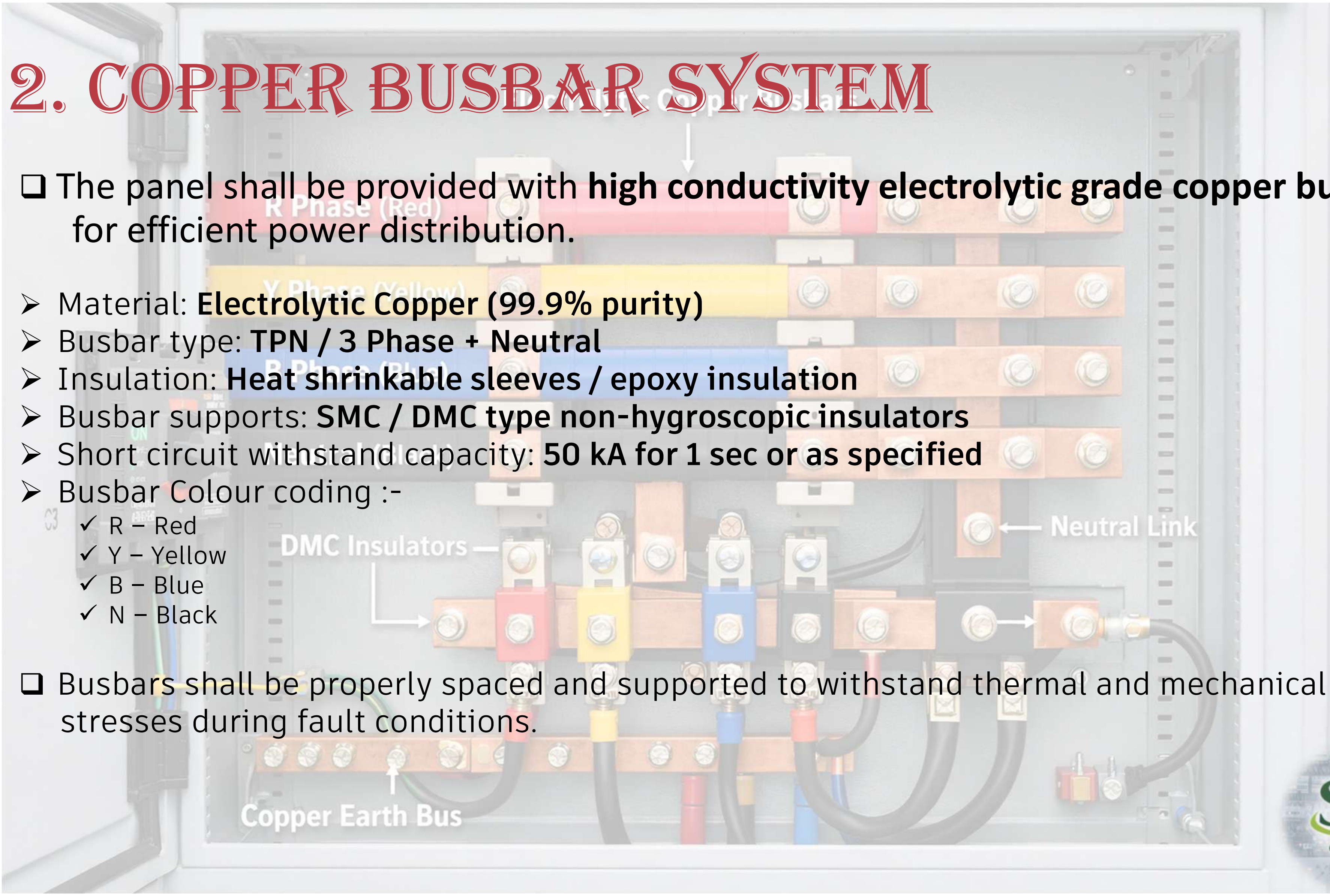
- Sheet thickness: **1.6 mm for structure, 1.6 mm for doors and covers**
- Floor mounted / wall mounted as required
- Powder coated with **epoxy polyester paint (minimum 7 tank process)**
- Colour shade: **RAL 7035 / Siemens Grey**
- Protection degree: **IP54**
- Adequate ventilation with louvers and filters
- Removable gland plates at bottom
- Fully compartmentalized structure Fabricated from **CRCA sheet steel**

2. Copper Busbar System

- ❑ The panel shall be provided with **high conductivity electrolytic grade copper busbars** for efficient power distribution.

- ➢ Material: **Electrolytic Copper (99.9% purity)**
- ➢ Busbar type: **TPN / 3 Phase + Neutral**
- ➢ Insulation: **Heat shrinkable sleeves / epoxy insulation**
- ➢ Busbar supports: **SMC / DMC type non-hygroscopic insulators**
- ➢ Short circuit withstand capacity: **50 kA for 1 sec or as specified**
- ➢ Busbar Colour coding :-
  - ✓ R – Red
  - ✓ Y – Yellow
  - ✓ B – Blue
  - ✓ N – Black

- ❑ Busbars shall be properly spaced and supported to withstand thermal and mechanical stresses during fault conditions.



3. INCOMING PROTECTION

❑ The panel shall be provided with suitable incoming protection such as :-

- **MCCB / ACB as required**
- Breaking capacity: **50kA or above**
- Adjustable overload and short circuit protection
- Rotary handle operation with door interlocking
- Mechanical ON/OFF indicator

4. OUTGOING FEEDERS

❑ Outgoing feeders shall be provided for EV chargers and auxiliary loads. :-

➢ TYPICAL CONFIGURATION :-

✓ MCCB / MCB outgoing feeders
✓ Individual protection for each EV charger circuit
✓ Proper cable termination points
✓ Phase indication lamps

5. METERING & INDICATION

- The panel shall include metering instruments for monitoring electrical parameters :-

  - **Digital multifunction meter**
  - Voltmeter with selector switch
  - Ammeter with selector switch
  - Current transformers (CTs)
  - LED indication lamps for RYB phases

- Measured parameters may include:

  - Voltage
  - Current
  - Power
  - Frequency
  - Power factor
  - Energy consumption

6. INTERNAL WIRING

- Control wiring: **FRLS copper wires**
- Ferruled at both ends
- Neatly dressed in PVC channels
- Proper numbering and identification
- Terminal blocks for all connections

7. EARTHING ARRANGEMENT

❑ THE PANEL SHALL BE PROVIDED WITH :-

- **COPPER EARTH BUS**
- TWO SEPARATE EARTHING STUDS
- PROPER EARTHING OF ALL METALLIC PARTS

8. CABLE ENTRY

- BOTTOM CABLE ENTRY THROUGH REMOVABLE GLAND PLATE
- SUITABLE **DOUBLE COMPRESSION CABLE GLANDS**
- ALUMINIUM / COPPER LUGS FOR TERMINATION

9. TESTING & STANDARDS

TESTING :-

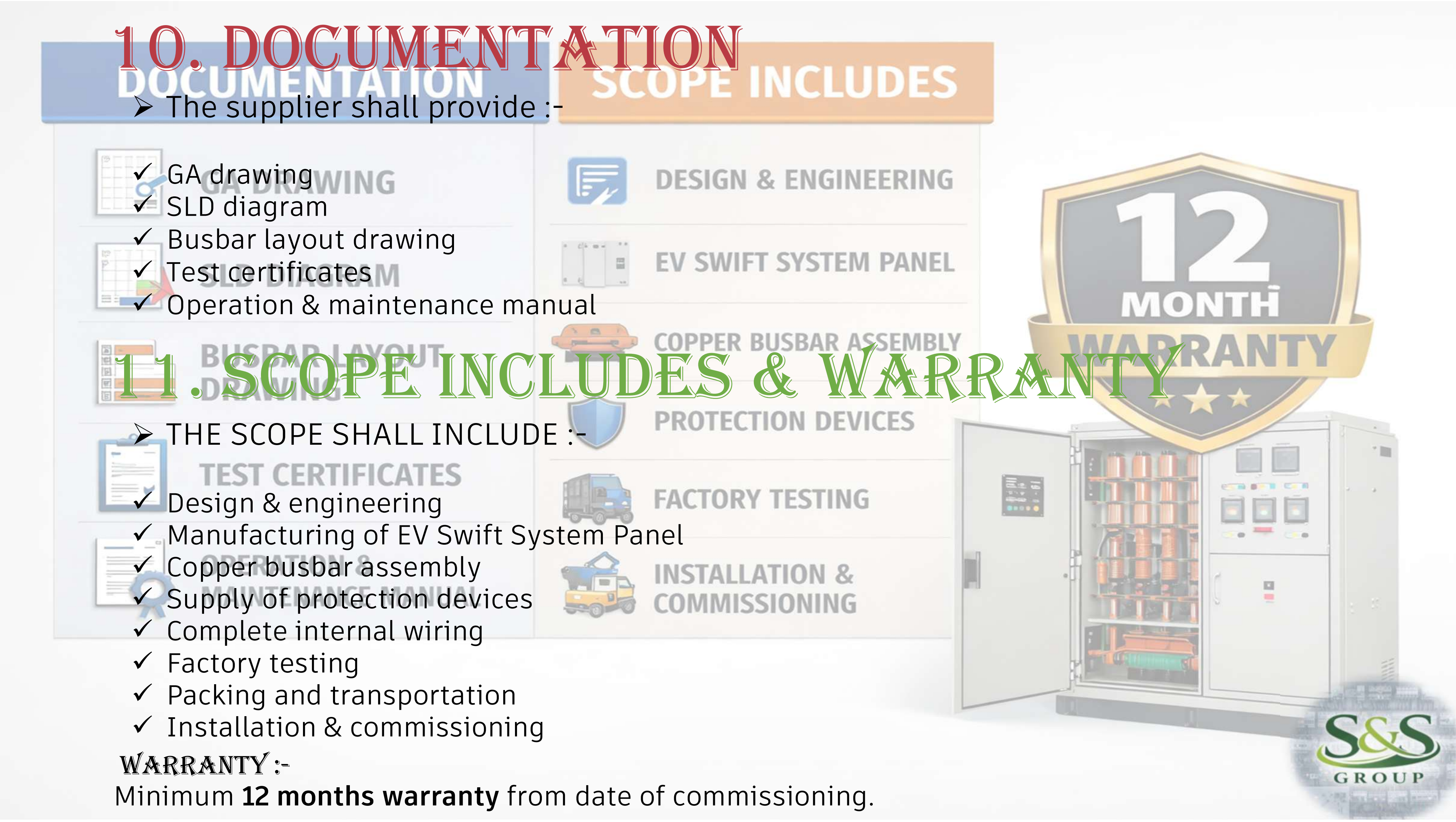
- The panel shall be tested as per **IS / IEC standards** before dispatch.
- **ROUTINE TESTS INCLUDE :-**

✓ Insulation resistance test
✓ High voltage test
✓ Functional test
✓ Busbar continuity test
✓ Protection device testing

Standards :-

- The panel shall conform to the following standards :-

✓ **IS 8623 / IEC 61439 – LT Panel Standard**
✓ **IS 5082 – Busbar Standard**
✓ **IS 375 – Marking & Identification**
✓ **IS 1248 – Indicating Instruments**



10. DOCUMENTATION

- The supplier shall provide :-

  - ✓ GA drawing
  - ✓ SLD diagram
  - ✓ Busbar layout drawing
  - ✓ Test certificates
  - ✓ Operation & maintenance manual

11 . SCOPE INCLUDES & WARRANTY

- THE SCOPE SHALL INCLUDE :-

  - ✓ Design & engineering
  - ✓ Manufacturing of EV Swift System Panel
  - ✓ Copper busbar assembly
  - ✓ Supply of protection devices
  - ✓ Complete internal wiring
  - ✓ Factory testing
  - ✓ Packing and transportation
  - ✓ Installation & commissioning

WARRANTY :-

Minimum **12 months warranty** from date of commissioning.